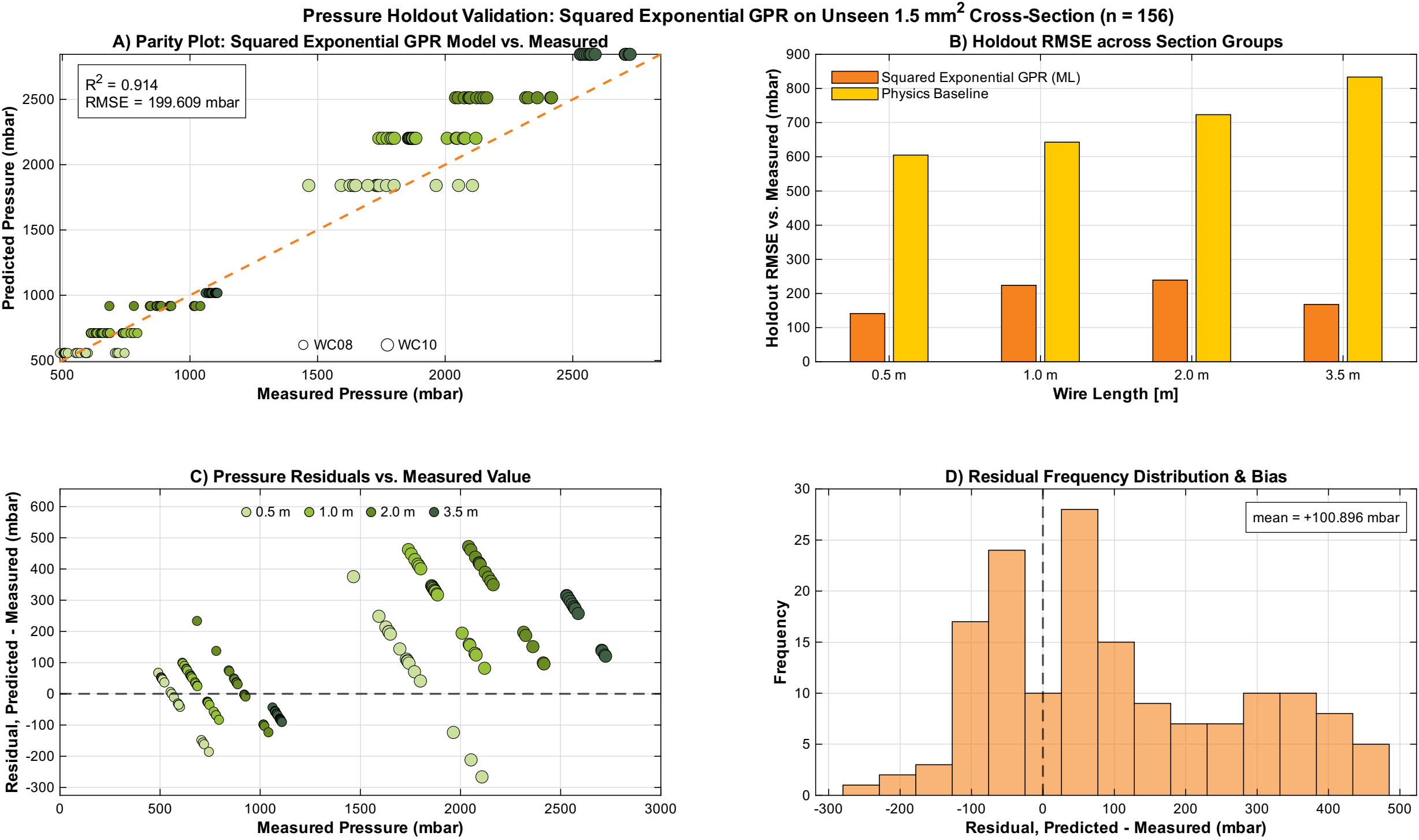

Pressure Holdout Validation: Squared Exponential GPR on Unseen 1.5 mm² Cross-Section (n = 156)
A) Parity Plot: Squared Exponential GPR Model vs. Measured
R² = 0.914
RMSE = 199.609 mbar
WC08
WC10
Predicted Pressure (mbar)
Measured Pressure (mbar)
B) Holdout RMSE across Section Groups
Squared Exponential GPR (ML)
Physics Baseline
Holdout RMSE vs. Measured (mbar)
0.5 m
1.0 m
2.0 m
3.5 m
Wire Length [m]
C) Pressure Residuals vs. Measured Value
0.5 m
1.0 m
2.0 m
3.5 m
Residual, Predicted - Measured (mbar)
Measured Pressure (mbar)
D) Residual Frequency Distribution & Bias
mean = +100.896 mbar
Frequency
Residual, Predicted - Measured (mbar)